

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

JetBrains annual survey of **23,262 developers** worldwide — a data-driven snapshot of programming language usage, tooling preferences, AI sentiment, and career trends.

2017–2024

Longitudinal language trend window

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Programming, scripting, and markup languages tracked

2025 planning

Implications for tech strategy and hiring priorities

Python Closes the Gap: 8 Years of Language Usage Shifts

Usage share (%) from 2017 to 2024. Top landscape is reshuffling: JavaScript down from peak, Python up sharply, TypeScript tripling, PHP declining.

Language	2017	2020	2022	2024	Trend
JavaScript	65%	70%	65%	61%	Declining from peak
Python	32%	55%	53%	57%	Steady growth
HTML/CSS	60%	61%	54%	51%	Gradual decline
SQL	42%	56%	49%	48%	Plateaued
Java	47%	54%	48%	46%	Slow decline
TypeScript	12%	28%	34%	37%	Strong growth
Go	8%	19%	19%	18%	Plateaued after growth
Rust	--	7%	9%	11%	Steady growth
PHP	30%	27%	20%	17%	Significant decline
Kotlin	2%	17%	16%	14%	Plateaued

Python Is Now the #1 Primary Language at 35%

Primary language share indicates installed base; adoption intent remains below 6% across languages, signaling ecosystem lock-in.



Adoption intent (plan to adopt as primary)	Python	TypeScript	Java	SQL
Share	6%	5%	4%	4%

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Top-10 developer satisfaction scores (horizontal bars). Type-safe, modern-tooling ecosystems dominate the leaderboard.



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Domain specialization matrix: language choice follows workload requirements rather than broad popularity.

Application Domain	Leading Language	Share	Why it matters
AI / Machine Learning	Python	31%	Core talent and ecosystem for model development
Data Science / Analytics	Python	28%	Default for notebooks, pipelines, experimentation
Data Processing & Analytics	Python	17%	Strong library depth for ETL and analysis
User Interface	TypeScript	62%	Front-end standardization and safer large codebases
User Interface	JavaScript	58%	Persistent runtime ubiquity across web stacks
APIs & Services	Java	49%	Mature enterprise backend footprint
Application Logic & Workflows	Java	55%	Stability and scalability in core business services
System Tools & Components	C++	27%	High-performance control and low-level access
Integrating with APIs & Services	Python	38%	Rapid integration and automation workflows

What This Means for Your Technology Strategy in 2025

- **Prioritize Python hiring for AI/ML and data teams:** it leads AI/ML (31%), data science (28%), and data integration-heavy workflows.
- **Standardize on TypeScript for frontend/full-stack paths:** strong growth (12% → 37% usage since 2017) and highest satisfaction score (199).
- **Build infrastructure tracks around Rust and Go:** both are top adoption targets; Rust usage climbs to 11% while Go remains strong at 18%.
- **Manage legacy risk in PHP/Ruby-heavy portfolios:** PHP falls from 30% to 17%; Ruby from 10% to 4%, implying tighter long-term talent pools.
- **Treat developer experience as retention strategy:** top-loved languages emphasize type safety and modern tooling, correlating with higher satisfaction.

Bottom line: choose language strategy by domain fit + developer sentiment, then align hiring pipelines to where momentum is accelerating.